Proof: ($\Rightarrow$) For $1 \le j \le n$,

$$A(Pe_j) = PDe_j.$$

$$Av_j = P(De_j) = P\begin{pmatrix} 0 \\ \vdots \\ \lambda_j \\ \vdots \\ 0 \end{pmatrix} = \lambda_j v_j.$$

($\Leftarrow$) go reverse.

→ Diagonalizability of A $\Leftrightarrow$ existence of a basis for $\mathbb{C}^n$ consisting of eigenvectors of A.

→ For $A \in M_n(\mathbb{C})$, Suppose $\sigma(A)$ contains 'n' distinct eigenvalues.
Let $\sigma(A) = \{\lambda_1, \lambda_2, \ldots, \lambda_n\}$, $|\sigma(A)| = n$

$\Leftrightarrow$ The algebraic multiplicity of each eigenvalue of A equals the geometric multiplicity of it.

ex:-1) $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

A A is already diagonalized.
eigenvalues are 1, 0.

$\sigma(B) = \{1\}$.
~~ateg~~ algebraic multiplicity of $1 = 2$.

$E_1 = \{x \in \mathbb{C}^2 \mid \cancel{A}Bx = x\}$.

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\begin{bmatrix} x+y \\ y \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}.$$

$\therefore$ $x \in \mathbb{C}$ $y = 0.$

$x \in$ span $\{(1,0)\}$.

$\therefore$ geometric multiplicity $= 1$. of evalue 1

AM $\neq$ GM.

$\therefore$ Not diagonalizable.

2) $A = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 3 & -2 \\ 2 & 1 & 0 \end{bmatrix}$ Identify if it is diagonalizable. If yes, find P, D.

A $\chi_A(t) = \det(tI_3 - A)$

$$= \begin{vmatrix} t-1 & 0 & 0 \\ -4 & t-3 & +2 \\ -2 & -1 & t \end{vmatrix}$$

$$= (t-1)[t^2 - 3t + 2]$$

$$= (t-1)^2(t-2). \quad \Rightarrow \sigma(A) = \{1, 1, 2\}$$

$\lambda = 1 \Rightarrow$ AM $= 2$.

GM $\Rightarrow$

$\lambda = 2 \Rightarrow$ AM $= 1$.

$\Rightarrow$ GM $= 1$ (has to be).

AM = GM.

$Ax = \cancel{2}x.$

$$\begin{bmatrix} 1 & 0 & 0 \\ 4 & 3 & -2 \\ 2 & 1 & 0 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2x \\ 2y \\ 2z \end{bmatrix}$$

$$\begin{bmatrix} x \\ 4x+3y-2z \\ 2x+y \end{bmatrix} = \begin{bmatrix} 2x \\ 2y \\ 2z \end{bmatrix}$$

$\cancel{x=0}$. $y=2z$.

$\therefore \cancel{(x,y,z) = (0, 2z, z)}$.

$\cancel{\text{span}\{(0,2,1)\}}$.

$4x + 2y - 2z = 0$

$2x + y - z = 0$.

$z = 2x+y$.

$\therefore (x,y,z) = (x, y, 2x+y)$.

$E_1 = \text{span}\{(1,0,2), (0,1,1)\}$.

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$\Rightarrow$ Theorem:- Let $A \in M_n(\mathbb{C})$ & $\lambda \in \sigma(A)$ then algebraic multiplicity of $\lambda \geq$ geometric multiplicity of λ $(m \geq k)$.

Proof:- Since the $\dim(E_\lambda) = k$, one can choose a basis for E_λ say $\{x_1, \ldots x_k\}$ s.t. $Ax_i = \lambda x_i \; \forall \; 1 \leq i \leq k$.

Extend to a basis $\mathcal{B} = \{x_1, \ldots x_k, x_{k+1}, \ldots x_n\}$ for $\mathbb{C}^n$

$$[A]_{\mathcal{B}} = \left(\begin{array}{c|c} \begin{pmatrix} \lambda & & \\ & \ddots & \\ & & \lambda \end{pmatrix}_{k \times k} & * \\ \hline 0 & W \end{array}\right)$$

characteristic polynomial of A / $[A]_\beta$ $= (t-\lambda)^k \, \chi_W(t)$

$$\| \\ (t-\lambda)^m q(t) \\ (q(\lambda) \neq 0)$$

This implies $m \geq k$.

→ A is diagonalizable then ($\Longleftrightarrow$)

$$\mathbb{C}^n = \bigoplus_{i=1}^{k} E_{\lambda_i} \quad \text{(direct sum)}.$$

$\lambda_1, \ldots, \lambda_k$ are the distinct eigenvalues of A.

⇒ Projection:

→ V is ips, $\dim(V) = n$, $W \leq V$, $\{w_1, \ldots, w_k\}$ is an orthonormal basis for W.

Given $x \in V$; $\exists !\, u \in W$ s.t.

$$\|x-u\| \leq \|x-w\| \quad \forall \, w \in W.$$

Description of $u = \sum_{i=1}^{k} \langle x, w_i \rangle w_i$

$$= \mathrm{Proj}_W(x)$$

ex:- 1) $W = \text{span}\left\{\begin{pmatrix}1\\0\\-1\end{pmatrix}, \begin{pmatrix}1\\1\\1\end{pmatrix}\right\}$ $x = \begin{pmatrix}1\\2\\3\end{pmatrix}$

$v \in \mathbb{R}^3$.

Compute $\text{Proj}_W(x)$.

A $W = \text{span}\left(\left\{\begin{matrix}1/\sqrt{2}\\0\\-1/\sqrt{2}\end{matrix}\right\}, \left\{\begin{matrix}1/\sqrt{3}\\1/\sqrt{3}\\1/\sqrt{3}\end{matrix}\right\}\right)$

$\text{Proj}_W(x) = \langle x, w_1\rangle w_1 + \langle x, w_2\rangle w_2$.

$= \begin{pmatrix}-1\\0\\1\end{pmatrix} + \begin{pmatrix}2\\2\\2\end{pmatrix}$

$= \begin{pmatrix}1\\2\\3\end{pmatrix}$

Let $P = [w_1 | w_2] = \begin{bmatrix}1/\sqrt{2} & 1/\sqrt{3}\\0 & 1/\sqrt{3}\\-1/\sqrt{2} & 1/\sqrt{3}\end{bmatrix}$

Projection matrix.

$PP^* x = \begin{bmatrix}1/\sqrt{2} & 1/\sqrt{3}\\0 & 1/\sqrt{3}\\-1/\sqrt{2} & 1/\sqrt{3}\end{bmatrix}\begin{bmatrix}1/\sqrt{2} & 0 & -1/\sqrt{2}\\1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3}\end{bmatrix} x$.

$= \begin{bmatrix}5/6 & 1/3 & -1/6\\1/3 & 1/3 & 1/3\\-1/6 & 1/3 & 5/6\end{bmatrix}\begin{bmatrix}x\\y\\z\end{bmatrix}$.

$PP^* x = \text{Proj}_W(x)$.